

AI/NLP Risk Engine

Turning unstructured financial text into traceable, structured risk signals

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Prototype focus: ingestion → normalization → sentiment and event heuristics → impact scoring → structured output → portfolio stress scenarios

A reproducible prototype built around explicit contracts, source traceability, local persistence and transparent assumptions.

1. Problem & objective

Problem. Financial information appears in different formats and channels. Comparing records manually makes it difficult to consistently identify sentiment, event type and potential risk relevance.

Objective. Normalize financial text into a machine-readable signal with source metadata, sentiment, event category, impact score and evidence/limitations, then demonstrate a downstream portfolio stress workflow.

Input challenge	Engine response	Output value
Different source schemas	Configurable adapters + normalized contract	Consistent record fields
Ambiguous language	Deterministic phrase/keyword baseline	Sentiment + event label
Risk is multidimensional	Documented weighted heuristic	Impact score 1–10 + components
Hard to audit downstream output	IDs, timestamp and source URL retained	Traceable JSON + SQLite

Design principle: make assumptions and limitations visible instead of presenting heuristic scores as certain predictions.

2. Architecture & data flow

The system is organized as small, testable stages. Adapters normalize input; the analyzer produces a RiskSignal; the validated output can be persisted and consumed by the stress-test module.

1. Sources	2. Ingestion	3. Normalize	4. Risk analysis	5. Validate & store	6. Downstream
GDELT headline API Local financial-text CSV Synthetic fixtures	Source adapters Timestamp parsing Source IDs / URLs	Required fields Deduplication Metadata preservation	Sentiment lexicon Event keyword taxonomy Weighted impact score	RiskSignal contract JSON outputs SQLite tables	Synthetic portfolio Event-sector shock rules Scenario report

Implementation: Python · standard-library CSV/JSON/SQLite · configurable source adapters · unittest regression suite

GDELT live fetch is optional; a previous attempt encountered HTTP 429 rate limiting. The default demo is deterministic and does not depend on live API access.

3. RiskSignal contract & explainability

Every input is mapped to a structured signal so downstream consumers do not need to parse free text again.

Field group	What it captures	Why it matters
Identity & provenance	signal_id, record_id, timestamp, source type/ID, URL	Reproducibility and source-to-signal traceability
Entities	Candidate organizations/tickers from available metadata and heuristics	Potentially connects text to affected entities; extraction needs improvement
Sentiment	Score in [-1, +1], label, heuristic confidence/method	Standardized polarity signal, not a calibrated probability
Event	Controlled event taxonomy + heuristic confidence/method	Makes event types queryable; keyword false positives are possible
Impact & evidence	Score 1–10, components, weights, matched phrases, limitations	Transparent baseline and audit trail

Key design choice: sentiment, event classification and impact are separate outputs. Negative sentiment alone should not determine financial impact.

4. Impact-scoring methodology

The current impact score is an expert-weighted, rule-based baseline. The weights are documented assumptions—not weights learned from market outcomes.

Component	Weight	Interpretation
Event severity	35%	Baseline severity assigned to the event category
Business/financial consequence	25%	Heuristic estimate of consequence
Scope	20%	Heuristic breadth of affected scope
Entity relevance	10%	Whether a relevant entity/ticker is identified
Sentiment magnitude	10%	Absolute sentiment score; polarity handled separately

$$R = 0.35E + 0.25C + 0.20S + 0.10N + 0.10M$$

Impact = clip(round($1 + 9R$), 1, 10), where each component is normalized to [0,1] and M is absolute sentiment magnitude.

Interpretation: a transparent prioritization heuristic, not probability of loss, price direction, VaR, or an empirically validated risk forecast.

5. Data sources & handling

Source	Role in prototype	Caveat
Financial Tweets CSV (historical)	Second-source/local CSV adapter test; 28,275 unique records after deduplication	Historical data, not live sentiment; redistribution terms must be respected
GDELT DOC API	Optional live headline adapter	Fetch attempt returned HTTP 429; no claim of successful live ingestion in this run
Synthetic JSON fixtures	Deterministic regression tests and demo	Invented examples, not real market events
FinancialPhraseBank	Candidate source for separate sentiment validation	Not claimed as integrated or evaluated in the current run

The CSV experiment generated structured outputs and SQLite records. It is a processing demonstration—not evidence that the labels are correct.

6. Prototype results

28,275	17 / 17	6
Unique historical CSV records processed	Automated tests passing	Synthetic demo scenarios

Observed output distribution from the historical CSV run

Signal field	Observed result	Interpretation
Event type	24,720 classified as OTHER	High default/unknown rate; taxonomy needs refinement
Sentiment	27,533 neutral; 690 negative; 52 positive	Likely baseline/lexicon bias; not a ground-truth distribution
Impact	24,585 assigned score 4	Heuristic default concentration; not measured market risk

Test success validates expected behavior on controlled fixtures and input contracts; it does not measure real-world NLP accuracy.

7. Downstream: synthetic portfolio stress testing

Risk signals are passed to a portfolio module that applies event-category sector shocks, scales them by the impact score, and computes illustrative post-scenario values.

Synthetic event	Impact	Portfolio change	Status
Credit event	8	−2.88%	Mapped
Earnings	5	−2.45%	Mapped
Regulatory	7	−2.20%	Mapped
Product	5	−1.11%	Mapped
Operational	7	−1.71%	Mapped
Other	3	0.00%	Unmapped

Baseline portfolio: 6,000,000 arbitrary units. The `OTHER` zero-change result means no sector mapping is defined—not that the event has no risk.

The shocks are illustrative adverse-scenario assumptions. They are not based on estimated market betas, historical event studies, or observed portfolio returns.

8. Limitations & next steps

Current limitation	Responsible next step
Keyword event classification; high OTHER rate	Build a labelled validation set; measure per-class precision/recall and confusion matrix
Phrase-lexicon sentiment; neutral-heavy outputs	Compare against a permitted financial sentiment dataset; report macro-F1 and class metrics
Heuristic entity candidates can be false positives	Improve entity linking and ticker validation; retain evidence spans
Impact weights are expert assumptions	Conduct sensitivity analysis and calibrate against defensible outcome labels
Stress shocks are illustrative	Use documented historical scenarios and validated exposures before any real-world use

No trading or investment use is intended. Real deployment requires validated data access, evaluation, calibration, monitoring and governance.

9. Demo walkthrough & impact

Suggested 4–5 minute walkthrough

Time	Show	Explain
0:00–0:30	Problem + architecture diagram	Convert multi-source text into structured, traceable risk signals
0:30–1:15	Run <code>`python run_demo.py`</code>	Deterministic six-record replay; no API credentials needed
1:15–2:00	Open <code>`data/sample/risk_signals.json`</code>	Show provenance, sentiment, event, impact components and limitations
2:00–3:00	Show CSV command and SQLite persistence summary	28,275 unique historical records processed; clarify heuristic outputs
3:00–4:15	Open <code>`data/sample/stress_test_results.json`</code>	Explain mapped shocks and explicitly unmapped OTHER event
4:15–5:00	Run test suite; summarize limits	17 tests pass; no claim of measured model accuracy

Domain impact: the prototype demonstrates a transparent path from unstructured financial text to queryable risk signals and scenario analysis. Its immediate value is workflow standardization and traceability—not automated investment decisions.

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